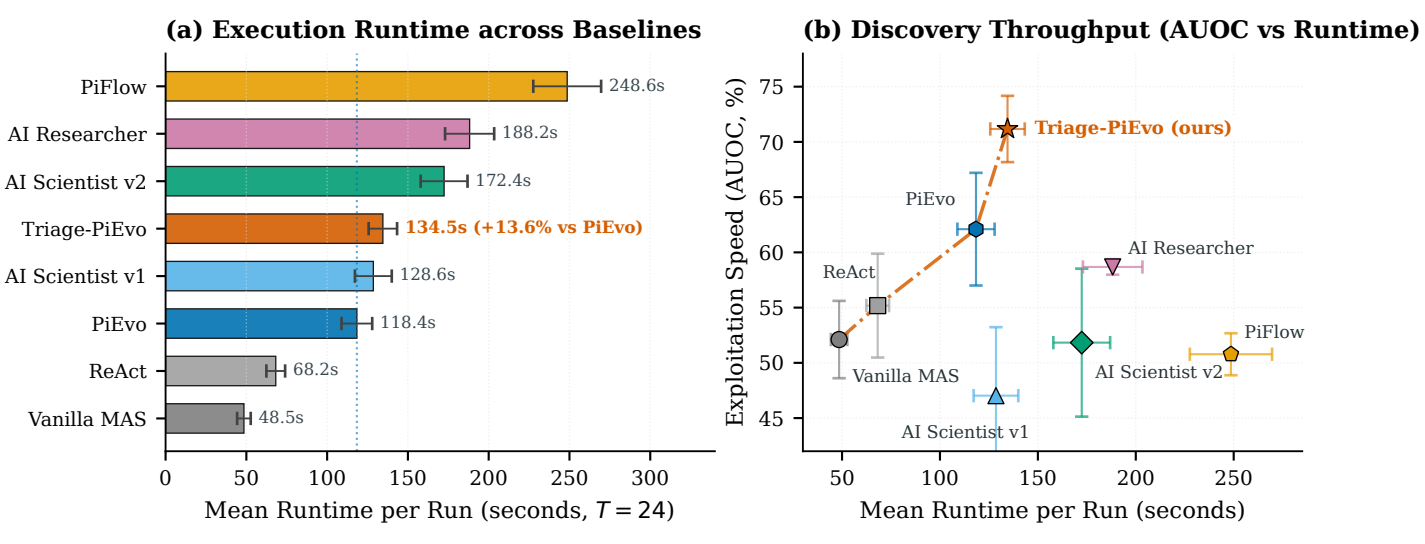
(a) Execution Runtime across Baselines
PiFlow
AI Researcher
AI Scientist v2
Triage-PiEvo
AI Scientist v1
PiEvo
ReAct
Vanilla MAS
248.6s
188.2s
172.4s
134.5s (+13.6% vs PiEvo)
128.6s
118.4s
68.2s
48.5s
0
50
100
150
200
250
300
Mean Runtime per Run (seconds, $T = 24$)
(b) Discovery Throughput (AUOC vs Runtime)
Exploitation Speed (AUOC, %)
75
70
65
60
55
50
45
Triage-PiEvo (ours)
PiEvo
ReAct
Vanilla MAS
AI Researcher
AI Scientist v2
PiFlow
AI Scientist v1
50
100
150
200
250
Mean Runtime per Run (seconds)